

A yellow paper sculpture of a jagged line graph, resembling a sawtooth pattern, is positioned on a light gray surface. The word "DEPRECIATION" is overlaid in white, bold, sans-serif capital letters across the center of the sculpture. The sculpture is made of a single strip of yellow paper, folded to create sharp peaks and valleys. The word is centered horizontally and partially obscures the middle of the sculpture. The background is a plain, light gray gradient.

DEPRECIATION

INTRODUCTION

- Any equipment which is purchased today will not work for ever. This may be due to wear and tear of the equipment or obsolescence of technology.
- Hence, it is to be replaced at the proper time for continuance of any business.
- The replacement of the equipment at the end of its life involves money.
- This must be internally generated from the earnings of the equipment.
- The recovery of money from the earnings of an equipment for its replacement purpose is called *depreciation fund* since we make an assumption that the value of the equipment decreases with the passage of time.
- Thus, the word “depreciation” means *decrease* in value of any physical asset with the passage of time.



METHODS OF DEPRECIATION



There are several methods of accounting depreciation fund. These are as follows:



1. Straight line method of depreciation



2. Declining balance method of depreciation



3. Sum of the years—digits method of depreciation



4. Sinking-fund method of depreciation

Straight Line Method of Depreciation

- In this method of depreciation, a fixed sum is charged as the depreciation amount throughout the lifetime of an asset such that the accumulated sum at the end of the life of the asset is exactly equal to the purchase value of the asset.
- Here, we make an important assumption that inflation is absent.

The formulae for depreciation and book value are as follows:

$$D_t = (P - F)/n$$

$$B_t = B_{t-1} - D_t = P - t \times [(P - F)/n]$$

Let

P = first cost of the asset,

F = salvage value of the asset,

n = life of the asset,

B_t = book value of the asset at the end of the period t ,

D_t = depreciation amount for the period t .

EXAMPLE 9.1 A company has purchased an equipment whose first cost is Rs. 1,00,000 with an estimated life of eight years. The estimated salvage value of the equipment at the end of its lifetime is Rs. 20,000. Determine the depreciation charge and book value at the end of various years using the straight line method of depreciation.

Solution

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$\begin{aligned} D_t &= (P - F)/n \\ &= (1,00,000 - 20,000)/8 \\ &= \text{Rs. } 10,000 \end{aligned}$$

In this method of depreciation, the value of D_t is the same for all the years. The calculations pertaining to B_t for different values of t are summarized in Table 9.1.

Table 9.1 D_t and B_t Values under Straight line Method of Depreciation

<i>End of year</i> (t)	<i>Depreciation</i> (D_t)	<i>Book value</i> ($B_t = B_{t-1} - D_t$)
0		1,00,000
1	10,000	90,000
2	10,000	80,000
3	10,000	70,000
4	10,000	60,000
5	10,000	50,000
6	10,000	40,000
7	10,000	30,000
8	10,000	20,000

EXAMPLE 9.2 Consider Example 9.1 and compute the depreciation and the book value for period 5.

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$D_5 = (P - F)/n$$

$$= (1,00,000 - 20,000)/8$$

$$= \text{Rs. } 10,000 \text{ (This is independent of the time period.)}$$

$$B_t = P - t \times (P - F)/n$$

$$B_5 = 1,00,000 - 5 \times (1,00,000 - 20,000)/8$$

$$= \text{Rs. } 50,000$$



Declining Balance Method of Depreciation

- In this method of depreciation, a constant percentage of the book value of the previous period of the asset will be charged as the depreciation amount for the current period.
 - This approach is a more realistic approach, since the depreciation charge decreases with the life of the asset which matches with the earning potential of the asset.
 - The book value at the end of the life of the asset may not be exactly equal to the salvage value of the asset. This is a major limitation of this approach.
-

The formulae for depreciation and book value are as follows:

$$D_t = K \times B_{t-1}$$

$$\begin{aligned} B_t &= B_{t-1} - D_t = B_{t-1} - K \times B_{t-1} \\ &= (1 - K) \times B_{t-1} \end{aligned}$$

The formulae for depreciation and book value in terms of P are as follows:

$$D_t = K(1 - K)^{t-1} \times P$$

$$B_t = (1 - K)^t \times P$$

Let

P = first cost of the asset,

F = salvage value of the asset,

n = life of the asset,

B_t = book value of the asset at the end of the period t ,

K = a fixed percentage, and

D_t = depreciation amount at the end of the period t .

Q2. Consider the following values and demonstrate the calculations of the declining balance method of depreciation by assuming 0.2 for K .

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$K = 0.2$$

The calculations pertaining to D_t and B_t for different values of t are summarized in Table using the following formulae:

$$D_t = K \times B_{t-1}$$

$$B_t = B_{t-1} - D_t$$

<i>End of year</i> (<i>n</i>)	<i>Depreciation</i> (<i>D_t</i>)	<i>Book value</i> (<i>B_t</i>)
0		1,00,000.00
1	20,000.00	80,000.00
2	16,000.00	64,000.00
3	12,800.00	51,200.00
4	10,240.00	40,960.00
5	8,192.00	32,768.00
6	6,553.60	26,214.40
7	5,242.88	20,971.52
8	4,194.30	16,777.22

If we are interested in computing D_t and B_t for a specific period t , the respective formulae can be used.

Consider previous values and calculate the depreciation and the book value for period 5 using the declining balance method of depreciation by assuming 0.2 for K .

Solution

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$K = 0.2$$

$$D_t = K(1 - K)^{t-1} \times P$$

$$\begin{aligned} D_5 &= 0.2(1 - 0.2)^4 \times 1,00,000 \\ &= \text{Rs. } 8,192 \end{aligned}$$

$$B_t = (1 - K)^t \times P$$

$$\begin{aligned} B_5 &= (1 - 0.2)^5 \times 1,00,000 \\ &= \text{Rs. } 32,768 \end{aligned}$$

Sum-of-the-Years-Digits Method of Depreciation

- In this method of depreciation also, it is assumed that the book value of the asset decreases at a decreasing rate. If the asset has a life of eight years, first the sum of the years is computed as
- Sum of the years = $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36 = n(n + 1)/2$
- The rate of depreciation charge for the first year is assumed as the highest and then it decreases. The rates of depreciation for the years 1–8, respectively are as follows: $8/36$, $7/36$, $6/36$, $5/36$, $4/36$, $3/36$, $2/36$, and $1/36$.

For any year, the depreciation is calculated by multiplying the corresponding rate of depreciation with $(P - F)$.

$$D_t = \text{Rate} \times (P - F)$$

$$B_t = B_{t-1} - D_t$$

The formulae for D_t and B_t for a specific year t are as follows:

$$D_t = \frac{n - t + 1}{n(n + 1)/2} (P - F)$$

$$B_t = (P - F) \frac{(n - t)}{n} \frac{(n - t + 1)}{(n + 1)} + F$$

Q3. Consider the given values and demonstrate the calculations of the sum-of-the-years-digits method of depreciation.

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$\text{Sum} = n(n + 1)/2 = 8 \times 9/2 = 36$$

The rates for years 1–8, are respectively $8/36$, $7/36$, $6/36$, $5/36$, $4/36$, $3/36$, $2/36$ and $1/36$.

The calculations of D_t and B_t for different values of t are summarized in Table 9.3 using the following formulae:

$$D_t = \text{Rate} \times (P - F)$$

$$B_t = B_{t-1} - D_t$$

Table 9.3 D_t and B_t under Sum-of-the-years-digits Method of Depreciation

<i>End of year</i> (n)	<i>Depreciation</i> (D_t)	<i>Book value</i> (B_t)
0		1,00,000.00
1	17,777.77	82,222.23
2	15,555.55	66,666.68
3	13,333.33	53,333.35
4	11,111.11	42,222.24
5	8,888.88	33,333.36
6	6,666.66	26,666.70
7	4,444.44	22,222.26
8	2,222.22	20,000.04

If we are interested in calculating D_t and B_t for a specific t , then the usage of the formulae would be better.

Consider the previous values and find the depreciation and book value for the 5th year using the sum-of-the-years-digits method of depreciation.

Solution

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$D_t = \frac{n - t + 1}{n(n + 1)/2} (P - F)$$

$$\begin{aligned} D_5 &= \frac{8 - 5 + 1}{8(8 + 1)/2} (1,00,000 - 20,000) \\ &= \text{Rs. } 8,888.88 \end{aligned}$$

$$B_t = (P - F) \frac{n - t}{n} \frac{n - t + 1}{n + 1} + F$$

$$\begin{aligned} B_5 &= (1,00,000 - 20,000) \frac{8 - 5}{8} \frac{8 - 5 + 1}{8 + 1} + 20,000 \\ &= 80,000 \times (3/8) \times (4/9) + 20,000 \\ &= \text{Rs. } 33,333.33 \end{aligned}$$

Sinking Fund Method of Depreciation

- In this method of depreciation, the book value decreases at increasing rates with respect to the life of the asset.

Let,

P = first cost of the asset,

F = salvage value of the asset,

n = life of the asset,

i = rate of return compounded annually,

A = the annual equivalent amount,

B_t = the book value of the asset at the end of the period t , and

D_t = the depreciation amount at the end of the period t .

The loss in value of the asset ($P - F$) is made available in the form of cumulative depreciation amount at the end of the life of the asset by setting up an equal depreciation amount (A) at the end of each period during the lifetime of the asset.

$$A = (P - F) \times [A/F, i, n]$$

The fixed sum depreciated at the end of every time period earns an interest at the rate of $i\%$ compounded annually, and hence the actual depreciation amount will be in the increasing manner with respect to the time period. A generalized formula for D_t is

$$D_t = (P - F) \times (A/F, i, n) \times (F/P, i, t - 1)$$

The formula to calculate the book value at the end of period t is

$$B_t = P - (P - F) (A/F, i, n) (F/A, i, t)$$

Consider Example 9.1 and give the calculations regarding the sinking fund method of depreciation with an interest rate of 12%, compounded annually.

Solution

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$i = 12\%$$

$$\begin{aligned} A &= (P - F) \times [A/F, 12\%, 8] \\ &= (1,00,000 - 20,000) \times 0.0813 \\ &= \text{Rs. } 6,504 \end{aligned}$$

Depreciation at the end of year 1 (D_1) = Rs. 6,504.

Depreciation at the end of year 2 (D_2) = $6,504 + 6,504 \times 0.12$
= Rs. 7,284.48

Depreciation at the end of the year 3 (D_3)
= $6,504 + (6,504 + 7,284.48) \times 0.12$
= Rs. 8,158.62

Depreciation at the end of year 4 (D_4)
= $6,504 + (6,504 + 7,284.48 + 8,158.62) \times 0.12$
= Rs. 9,137.65

<i>End of year</i> <i>t</i>	<i>Fixed</i> <i>depreciation</i> (Rs.)	<i>Net depreciation</i> D_t (Rs.)	<i>Book value</i> B_t (Rs.)
0	6,504	—	1,00,000.00
1	6,504	6,504.00	93,496.00
2	6,504	7,284.48	86,211.52
3	6,504	8,158.62	78,052.90
4	6,504	9,137.65	68,915.25
5	6,504	10,234.17	58,681.08
6	6,504	11,462.27	47,218.81
7	6,504	12,837.74	34,381.07
8	6,504	14,378.27	20,002.80
$B_t = B_{t-1} - D_t$			

Consider Example 9.1 and compute D_5 and B_7 using the sinking fund method of depreciation with an interest rate of 12%, compounded annually.

Solution

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$i = 12\%$$

$$D_t = (P - F) (A/F, i, n) (F/P, i, t - 1)$$

$$\begin{aligned} D_5 &= (P - F) (A/F, 12\%, 8) (F/P, 12\%, 4) \\ &= (1,00,000 - 20,000) \times 0.0813 \times 1.574 \\ &= \text{Rs. } 10,237.30 \end{aligned}$$

$$B_t = P - (P - F) (A/F, i, n) (F/A, i, t)$$

$$B_7 = P - (P - F) (A/F, 12\%, 8) (F/A, 12\%, 7)$$

$$= 1,00,000 - (1,00,000 - 20,000) \times 0.0813 \times 10.089$$

$$= 34,381.10$$